

Institutional Quantitative Research Report: Hybrid Volatility-Squeeze Engine & Execution Backtest Evaluation on Binance ETH/USDT

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Abstract

This report delivers an end-to-end evaluation of a high-frequency (10-minute bar) volatility forecasting and directional breakout trading engine applied to the **Binance ETH/USDT** market. The predictive model was calibrated and trained on historical data spanning **2024 through 2025** and strictly out-of-sample evaluated on the **first 6 months of 2026 (25,550 10-minute bars)**. We detail the architecture of a multimodal Deep Learning engine (GAF-CNN-LSTM) integrated with Stationary Wavelet Transforms (SWT) and Gramian Angular Summation Fields (GASF). We benchmark this engine against GARCH(1,1) and a blended Ensemble model, examining three execution variants. Notably, we highlight a critical empirical divergence: while the Ensemble model achieved statistical superiority in QLIKE loss on the test set, it failed to generate higher backtested net returns when paired with execution rules compared to the standalone Neural Network.

1 System Overview & Model Architecture

The predictive engine forecasts localized Garman-Klass volatility on the Binance ETH/USDT pair to identify high-probability "volatility squeezes" preceding explosive trend breakouts. The framework operates through a multi-stage data and spatial processing pipeline:

- Data Pipeline & Dataset Split:** Ingests 10-minute OHLCV market bars from Binance for ETH/USDT. The model is trained on full 2024–2025 historical data ($N \approx 105,120$ bars) and evaluated out-of-sample across H1 2026 ($N = 25,550$ test bars).
- Feature Engineering Layer:** Extracts a rolling window of $N = 32$ 10-minute bars across 6 stationary microstructural metrics: log returns ($\ln(C_t/C_{t-1})$), Garman-Klass volatility, log range ($\ln(H/L)$), volume acceleration, body momentum gap ($(C - O)/O$), and normalized close position ($(C - L)/(H - L)$).
- Wavelet Signal Decomposition:** Uses a Stationary Wavelet Transform (SWT) with Daubechies 4 (db4) wavelets to split raw ETH/USDT time series into *Approximation* (low-frequency trend) and *Detail* (high-frequency micro-flickers) sub-bands without phase shift.
- Spatial Encoding (GASF):** Converts 1D wavelet matrices into 2D pseudo-images using Gramian Angular Summation Fields. Values scaled to $[-1, 1]$ are converted to polar coordinates $\phi = \arccos(X)$, yielding a $32 \times 32 \times 12$ tensor:
$$\text{GASF}_{i,j} = \cos(\phi_i + \phi_j) \quad (1)$$
- Dual-Branch Neural Topology:** Branch A utilizes 2D Convolutional layers to extract spatial geometric features from the GASF maps. Branch B processes normalized continuous time series through an LSTM layer to retain sequential temporal context.
- Asymmetric Loss Optimization (QLIKE):** Trained using Quasi-Likelihood loss to heavily penalize volatility *underestimation* on ETH/USDT tail moves:

$$L_{\text{QLIKE}}(y, \hat{y}) = \frac{y}{\hat{y}} - \ln\left(\frac{y}{\hat{y}}\right) - 1 \quad (2)$$

2 Predictive Engine Benchmark & Ensemble Analysis

The Deep Learning architecture was benchmarked against a classical econometric GARCH(1,1) model and a weighted Ensemble model (50.7% NN / 49.3% GARCH) across the 25,550-bar H1 2026 test horizon on Binance ETH/USDT.

Table 1: Predictive Volatility Model Performance Metrics (Binance ETH/USDT, H1 2026 Test Set)

Model Architecture	Mean QLIKE Loss ($\downarrow$)	Volatility Correlation ($\uparrow$)
GARCH(1,1) Baseline	0.661454	0.6117
Hybrid GAF-CNN-LSTM NN	0.642872	0.6214
Ensemble (NN + GARCH)	0.620735	0.6506

2.1 Statistical vs. Backtest Alpha Divergence

A critical quantitative finding emerged during strategy backtesting on the 2026 test set: **although the Ensemble model yielded the lowest statistical QLIKE loss (0.6207) and the highest correlation (0.6506), it did not generate higher net trading returns when paired with directional breakout rules.**

The GARCH component smoothed out brief ETH volatility dips, introducing lag into rolling percentile rankings. Consequently, the Ensemble delayed gating entries during sudden compression breakdowns. The standalone Neural Network, optimized purely on spatial-temporal microstructures, generated sharper non-linear volatility drop predictions. This enabled precise trade entries at the exact moment of maximum price compression, yielding superior entry timing and higher risk-adjusted strategy returns (+14.06% vs. lower ensemble strategy returns).

3 Trading Strategy Execution Logic

The trading engine operates as a 10-minute volatility-squeeze breakout model governed by four execution parameters:

- **Entry Trigger:** ETH/USDT price closes outside a 60-bar Donchian channel, aligned with a 200-hour macro EMA filter and volume expansion.
- **Squeeze Gating:** Predicted volatility must fall in the bottom 30th percentile relative to a 432-bar rolling window.
- **Risk Management:** Initial hard stop set at 1.8%. A 50% partial take-profit triggers at +1.8%, ratcheting the trailing stop to breakeven.
- **Exit Rules:** Dynamic volatility-scaled target for the remaining runner, a trailing channel-midpoint exit after 1 hour, and a strict 12-hour max hold time limits trade duration.

4 Comparative Backtest Results (H1 2026 Evaluation)

We evaluated three operational variants of the execution engine over the H1 2026 out-of-sample test period on Binance ETH/USDT.

Table 2: Executive Strategy Backtest Performance Summary (Binance ETH/USDT H1 2026)

Performance Metric	Strategy 1: Clean Taker	Strategy 2: Post-Maker	Strategy 3: Stabilized
Total Net Return	+14.06%	+12.86%	+5.43%
Max Peak-to-Trough Drawdown	-10.00%	-7.94%	-3.95%
Annualized Sharpe Ratio	1.55	1.76	1.04
Profit Factor	1.08	1.11	1.09
Win Rate (%)	42.4%	43.2%	38.6%
Total Executed Trades	66	44	44
Total Exchange Fees Paid	8.22% (Taker 0.05%)	2.23% (Maker 0.02%)	3.85% (Taker 0.05%)
Primary Execution Mode	Taker Market Orders	Post-Only Limit	Adaptive Taker

4.1 Month-by-Month Net Return Breakdown (H1 2026 Out-of-Sample)

Table 3: Monthly Return Profiles for Strategy Variants (2026 Out-of-Sample)

Month	Strategy 1 (Clean Taker)	Strategy 2 (Post-Maker)	Strategy 3 (Stabilized)
Jan 2026	0.00%	0.00%	0.00%
Feb 2026	-2.32%	-3.41%	-1.81%
Mar 2026	+13.74%	+3.74%	-0.06%
Apr 2026	+0.33%	+3.56%	+3.01%
May 2026	+1.15%	+2.70%	+0.11%
Jun 2026	-7.81%	-6.43%	-2.98%
Jul 2026	+8.97%	+12.70%	+7.16%
Cumulative Total	+14.06%	+12.86%	+5.43%

5 Performance Analysis & Operational Takeaways

5.1 Strategy 1: Clean Baseline Taker (Commercial Recommendation)

Strategy 1 represents the most viable live execution model for Binance ETH/USDT. By using market taker orders (0.05% fee), it ensures 100% fill rates on explosive breakout candles. The strategy’s large winning trades easily absorb higher trading friction (8.22% cumulative fee drag), resulting in an optimal net return of **+14.06%**.

5.2 Strategy 2: Post-Only Maker Flaw (Adverse Selection)

While Strategy 2 shows reduced fee drag (2.23%) and lower backtested drawdown (-7.94%) on paper, it relies on the unfeasible assumption that passive limit orders fill at ETH breakout boundaries. In live Binance microstructure, **adverse selection** degrades performance: explosive winning breakouts gap away without filling resting limit orders, while false breakouts dump directly into resting bids, filling losing trades and missing outsized runners.

5.3 Strategy 3: Stabilized Anti-Chop Flaw (Expectancy Skew)

Strategy 3 attempts to curb drawdown by halving position sizes after a loss and adding a momentum-stall timer. Although this reduces max drawdown to -3.95%, it skews strategy expectancy: taking full-sized losses on false breakouts and catching subsequent macro ETH runners at half-size drops total net return to **+5.43%**.

6 Conclusion

The optimal deployment setup combines the standalone **GAF-CNN-LSTM Neural Network volatility squeeze gate** (trained on 2024–2025 Binance ETH/USDT data) with **Strategy 1 (Clean Baseline Taker Execution)**. This configuration prioritizes entry execution and non-linear risk timing over paper fee savings or overly defensive position scaling, delivering robust, real-world positive expectancy.